JOY STUDENT Assembly Guide

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Parts

The JOY STUDENT console combines an Arduino UNO, four Grove modules and an RTC module. All of them are required, the RTC included.

Component	Role	Connection
Arduino UNO R3 + Grove Base Shield	main board	USB-B, 5 V
Grove LCD RGB Backlight V4.0	16 × 2 display	I2C
Grove SHT31 or SHT40	temperature and humidity	I2C, address 0x44
RTC module DS1307 or DS3231	time, alarm, automatic night mode	I2C, address 0x68
Grove Thumb Joystick	menu navigation	A1 (X), A2 (Y)
Grove Buzzer	sounds and alerts	D3

The Grove cables already come with their connectors. You also need a 3D printer and a precision screwdriver for the small screws.

Screws

The build needs 25 screws of four types, plus 2 screws for the RTC. They bite directly into the PLA.

Screw	Qty	Use
M3 × 6 mm, pan head	4	Arduino UNO on the base (4 posts)
M3 × 12 mm, countersunk head	4	Base to shell (4 corners, from below)
M1.7 × 5 mm self-tapping, pan head	13	Grove modules: LCD (5), SHT31 or SHT40 (3), joystick (3), buzzer (2)
M1.7 × 5 mm self-tapping, pan head	2	RTC module (to be confirmed for your board)
M2 × 20 mm, pan head (4 mm max.)	4	Cover, tray and shell rim (4 corners)

Plan about 20 % extra screws: the M1.7 ones are tiny. The four tray screws are M2 × 20, not M2 × 6, because they are what holds the cover; for a test of the tray alone, M2 × 6 is enough.

Printing

Print the four parts in this order: base, tray, cover, shell.

1. **Base:** print it first to check that the UNO fits.
2. **Tray:** it carries the SHT and the joystick.
3. **Cover:** visible face against the build plate, posts pointing up, no supports. Its name is engraved 0.6 mm deep.
4. **Shell:** it receives the LCD.

The base and the shell are the same as for Desk Watch (same files).

Step-by-step assembly

The build takes seven steps; plug in the Grove cables before placing the tray.

1. **Base.** Place the UNO on its four posts and fix it with four M3 × 6 screws, without over-tightening. Fix the buzzer on one of the four 20 × 20 spots of the base with two M1.7 × 5 screws, then the RTC module on a free spot (its mounting depends on your board).
2. **Display.** Screw the LCD onto the five posts of the shell with five M1.7 × 5 screws. Plug in its Grove cable before closing.
3. **Tray, SHT.** Place the SHT module on its three posts, components facing up, and screw it down with three M1.7 × 5 screws.
4. **Tray, joystick.** Lay the joystick board flat, head facing up, Grove socket towards the back of the tray, and screw it down with three M1.7 × 5 screws without forcing. The socket and the cable loop stay exposed above the rear window.
5. **Cables.** LCD, SHT and RTC on I2C sockets, joystick on socket A1, buzzer on socket D3. Pass the plugs through the windows of the tray.
6. **Tray and cover.** Place the tray on the rim of the shell, then the cover on top: the four posts center themselves in the counterbores of the tray. Fit the four M2 × 20 screws in the corners. The 34 mm hole lets the joystick head through.
7. **Closing.** First test with the case still open (see the last section). Then bring the base up to the shell vertically, with the USB notch sliding down over the socket, and fix it with four countersunk M3 × 12 screws from below.

Wiring

All modules connect to the Base Shield with Grove cables. The complete wiring diagram is provided separately.

Module	Shield socket	Note
RGB LCD	I2C	Bus shared with the SHT and the RTC
SHT sensor	I2C	Shared bus
RTC	I2C	Shared bus, required
Joystick	A1	Yellow wire = X axis, white wire = button if your board exposes it
Buzzer	D3	Yellow wire = signal

Usual Grove cable colors (Seeed standard, to be confirmed on your cables): yellow = signal 1 (SCL, A1 or D3), white = signal 2 (SDA, A2 or D4, unused here), red = 5 V, black = GND.

The shield must be set to 5 V (VCC switch).

First start-up and checks

1. Install the “Grove - LCD RGB Backlight” library in the Arduino IDE.
2. Do not touch the joystick when powering on: its rest position is measured at start-up.
3. Upload the FR or EN sketch of the JOY STUDENT edition, with the case still open.
4. Check the display and the backlight.
5. Test the joystick: up = next, down = previous, right or click = confirm, left = back.
6. Check the temperature and humidity from the SHT, the time from the RTC and the sound of the buzzer.
7. Close the base only when everything works.